1. Why is a singly linked list suitable for the customer waiting list?

2. How does the prev reference help the music playlist?

3. When removing a middle song, which references must be updated, and why?

4. State the time complexity of adding a customer with a tail reference, serving a customer, cancelling by name, moving to the next song, and displaying all songs. Explain each answer.

5. Describe one bug you found during testing and how you fixed it.

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1. Singly linked list: It is suitable because customers are served in FIFO order (first in, first out), and new customers can be added at the end.

2. prev reference: It allows the playlist to move backward to the previous song.

3. Removing a middle song: Update the previous node's next and the next node's prev so the remaining songs stay connected.

4. Time complexity:

- Add customer: O(1) - the tail reference allows direct insertion at the end.

- Serve customer: O(1) - remove the head directly.

- Cancel by name: O(n) - search through the list to find the customer.

- Move to next song: O(1) - follow the next reference.

- Display all songs: O(n) - visit every song in the playlist.

5. Bug found: Removing the current song could leave current pointing to a deleted node. I fixed it by updating current to the next song, or the previous song if no next song exists.